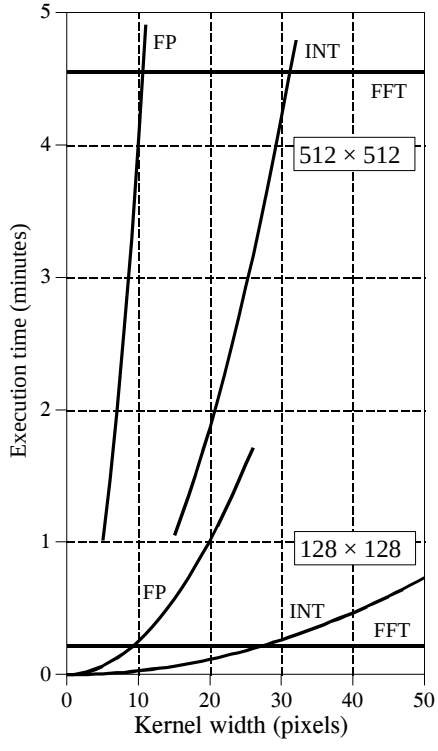

FIGURE 24-16
Execution time for image convolution. This graph shows the execution time on a 100 MHz Pentium processor for three image convolution methods: conventional convolution carried out with floating point math (FP), conventional convolution using integers (INT), and FFT convolution using floating point (FFT). The two sets of curves are for input image sizes of 512×512 and 128×128 pixels. Using FFT convolution, the time depends only on the image size, and not the size of the kernel. In contrast, conventional convolution depends on both the image and the kernel size.

in about 15 seconds using FFT convolution, while a 512×512 image requires more than 4 minutes. Adding up the number of calculations shows that the execution time for FFT convolution is proportional to $N^2 Log_2(N)$, for an $N \times N$ image. That is, a 512×512 image requires about 20 times as long as a 128×128 image.

Conventional convolution has an execution time proportional to $N^2 M^2$ for an $N \times N$ image convolved with an $M \times M$ kernel. This can be understood by examining the program in Table 24-1. In other words, the execution time for conventional convolution depends *very strongly* on the size of the kernel used. As shown in the graph, FFT convolution is faster than conventional convolution using floating point if the kernel is larger than about 10×10 pixels. In most cases, integers can be used for conventional convolution, increasing the break-even point to about 30×30 pixels. These break-even points depend slightly on the size of the image being convolved, as shown in the graph. The concept to remember is that FFT convolution is only useful for *large* filter kernels.